

Experiment 7.4

Determination and plotting of Capacitance vs Dielectric Constant for a Coaxial Cable

Code File Name: Exp7_4_Capacitance_vs_DielectricConstant_CoaxialCable.m

Code

```
clc;
epsilon0 = 8.854e-12;
a = 0.01;
b = 0.05;
er = linspace(1,5,100);
epsilon = epsilon0*er;
C = (2*pi*epsilon)./log(b/a);
plot(er,C,'LineWidth',2);
xlabel('Dielectric constant');
ylabel('Capacitance (F/m)');
title('Capacitance vs dielectric constant (coaxial cable)');
```

Output Graph

